CropMind

Home Dashboard Telemetry AI Advisor About Logout Light Theme

Start Crop Diagnosis

SELECT CROP TO INSPECT

Wheat (Rabi)

SOIL TYPE

Loamy

WATERING / IRRIGATION

Sparse / Rainfed

OBSERVED SYMPTOMS / FARM NOTES

rings, plant growth is stunted and flowers are dropping.

Analyze Health & Remedy

Advisor Core Idle

Fill out the diagnostic panel on the left to request a crop analysis. The AI agronomist will diagnose likely diseases and outline a detailed organic/chemical treatment checklist.

AI Assistant

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Running Diagnostics...

AI Diagnostics Running

Compiling Registered Crop parameters & history...

AI Assistant

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Analyze Health & Remedy

DIAGNOSIS REPORT

Start New Analysis

Severe Drought Stress leading to Nutrient Imbalance and Reproductive Failure

SEVERITY

HIGH

AI CONFIDENCE

95%

Suggested Treatment Plan

Mark tasks as checked as you complete them on your farm.

- [ ] Immediately provide adequate and consistent irrigation to relieve water stress. Prioritize critical stages (e.g., flowering).
- [ ] Monitor soil moisture levels carefully to ensure optimal water availability, avoiding both waterlogging and drought conditions.
- [ ] Once water stress is alleviated, consider a balanced foliar nutrient spray (e.g., NPK 19:19:19 or a micronutrient mix) to help plants recover and address any secondary nutrient deficiencies caused by poor uptake.

AI Assistant

Preventive Measures

- Implement a well-planned and consistent irrigation schedule for wheat, especially during critical growth stages like crown root initiation, tillering, jointing, flowering, and grain filling.
- Utilize soil moisture monitoring techniques (e.g., tensiometers, gravimetric methods, or even manual feel) to guide irrigation frequency and volume.
- Consider incorporating organic matter into the loamy soil to improve its water retention capacity and overall soil health.
- Explore water conservation practices such as mulching to reduce evaporation and promote better soil moisture retention.

NPK & Water Guidance

The sparse and irregular irrigation is the primary culprit. Wheat requires consistent moisture, especially during flowering, and irregular watering leads to severe stress, impacting nutrient uptake even if fertilizers are present in the soil. Immediately focus on establishing a regular and sufficient irrigation schedule. Once the plant recovers from water stress, assess nutrient status. If symptoms of specific nutrient deficiencies persist, apply appropriate fertilizers. For instance, if nitrogen deficiency is suspected, apply urea. However, ensure the soil has adequate moisture for nutrient uptake before applying granular fertilizers. A foliar spray might be beneficial for rapid nutrient assimilation under stress recovery.

CropMind AI - Backend Verification Logs

Vite Port 5173 | Express Port 5000

[HTTP] POST /api/ai/diagnose - Status: 200 OK

Headers: Authorization Bearer JWT Token

Response: { "diagnosis": "Severe Drought Stress...", "confidence": 95, ... }

Database Mode: In-Memory Fallback Developer Mock Store

Gemini Integration: Live API key verification complete